Assignment 1: Multi-Instance Object Recognition in Cluttered Scenes

1. Objective

The goal of this assignment is to detect and localise multiple instances of a known object in a single cluttered photograph. Instances may differ in scale and orientation and may be partly occluded by other objects. The pipeline uses OpenCV for image I/O and SIFT feature extraction. Feature matching, parameter-space clustering, geometric estimation, verification and duplicate suppression are implemented in the accompanying code.

2. Data

- **Template.** An isolated photograph of a Kurkure packet (`data/template.png`, 2591×3674).
- **Query scene.** A photograph of my lab desk (`data/scene.png`, 4032×3024) containing several Kurkure packets among other clutter, at different distances and orientations, with partial occlusion.



Figure 1. Left: isolated Kurkure template. Right: cluttered lab-desk query.

3. Recognition pipeline

3.1 SIFT features

SIFT keypoints and 128-D descriptors are extracted on both images (28,667 template keypoints, 38,802 scene keypoints). Descriptors are L1-normalised and square-rooted (RootSIFT) before matching so that Euclidean distance corresponds to the Hellinger distance between gradient histograms.

3.2 Feature correspondence

For each scene descriptor the two nearest template descriptors are found by explicit Euclidean distance. A match is kept only if it passes Lowe's ratio test $d1/d2 < \rho$ with $\rho = 0.80$. Matching is performed from the scene to the template so that each instance can accumulate its own correspondences rather than competing for a fixed budget of template features. This reduced 38,802 nearest-neighbour pairs to 2,842 matches (rejection rate 92.7%).



Figure 2. Naive nearest-neighbour matches without a ratio test. Correspondences spread across background clutter and do not identify object instances.



Figure 3. Matches remaining after Lowe's ratio test. Clutter is reduced, but the output is still an unordered set of correspondences.

3.3 4D generalised Hough transform

A single SIFT match implies a similarity pose (centre, scale, orientation). Each match therefore casts a vote in a four-dimensional parameter space (c_x, c_y, $\log_2 s$, $\Delta\theta$). Votes are broadened to neighbouring bins so that clusters straddling a bin boundary are not split. Bins with at least three votes form candidate object hypotheses. In this run the vote table contained 3,037 occupied bins, of which 1,154 cleared the vote threshold; the strongest bin held 1,031 votes.

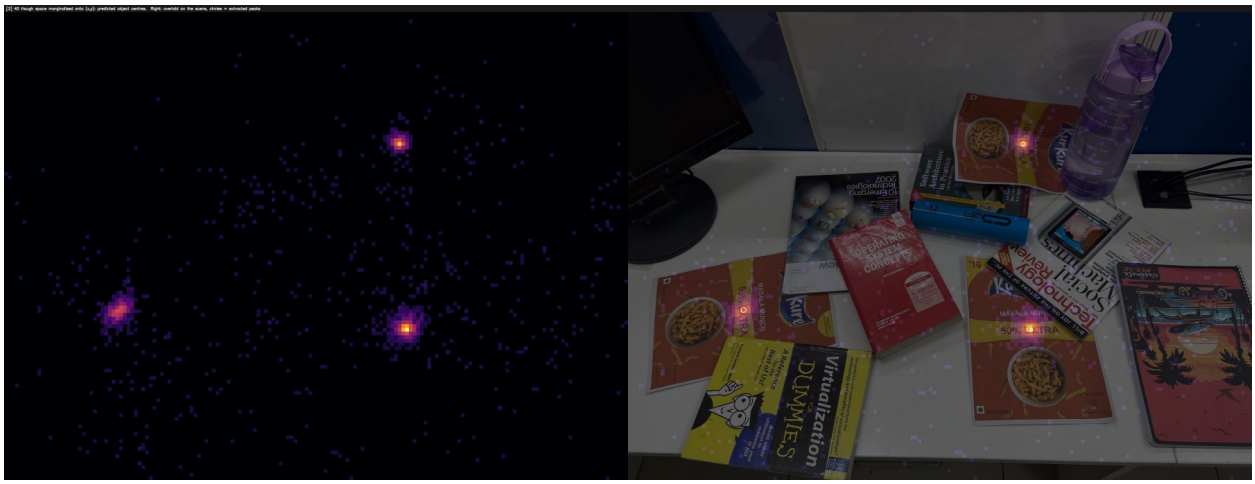


Figure 4. Hough votes marginalised onto the image plane. Peaks correspond to likely object centres.

3.4 Robust affine estimation

For each candidate bin a 2D affine transform is estimated with RANSAC. Minimal samples of three correspondences are solved exactly; surviving models are refined by least squares on the inlier set ($A\,x = b$). Degenerate samples are rejected by a collinearity check, and fitted transforms are discarded if they imply a reflection or excessive shear.

3.5 Iterative extraction

One affine model describes one instance. After a verified detection is emitted, its inlier matches are removed and the Hough table is rebuilt. The loop repeats until no remaining bin yields a verified fit. This sequential inlier subtraction recovers multiple instances without rediscovering the strongest one repeatedly.

3.6 Verification and non-maximum suppression

A hypothesis is accepted only if it has sufficient inliers, a coherent inlier ratio, acceptable mean reprojection error, geometric sanity, adequate inlier spread across the template, and a positive appearance check based on zero-mean normalised cross-correlation. Overlapping boxes that describe the same physical packet are merged with polygon IoU non-maximum suppression.

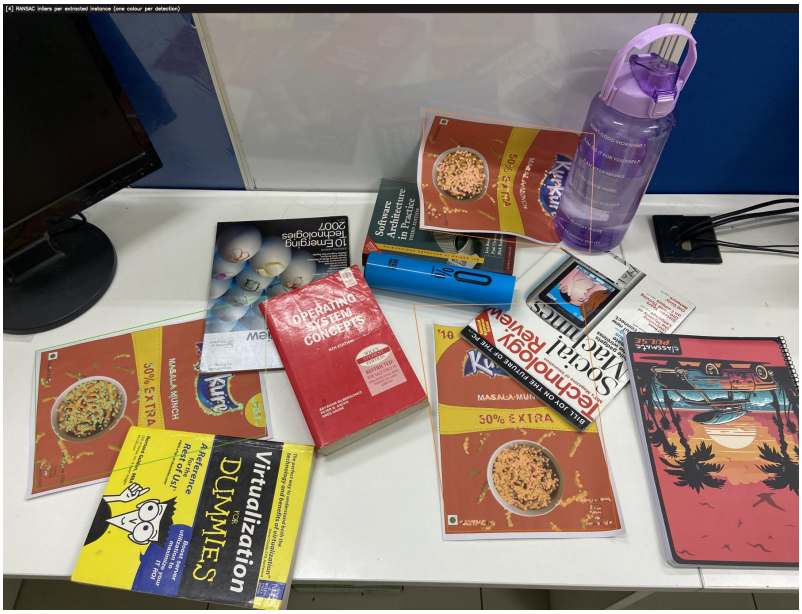


Figure 5. RANSAC inliers for each accepted instance after greedy extraction.

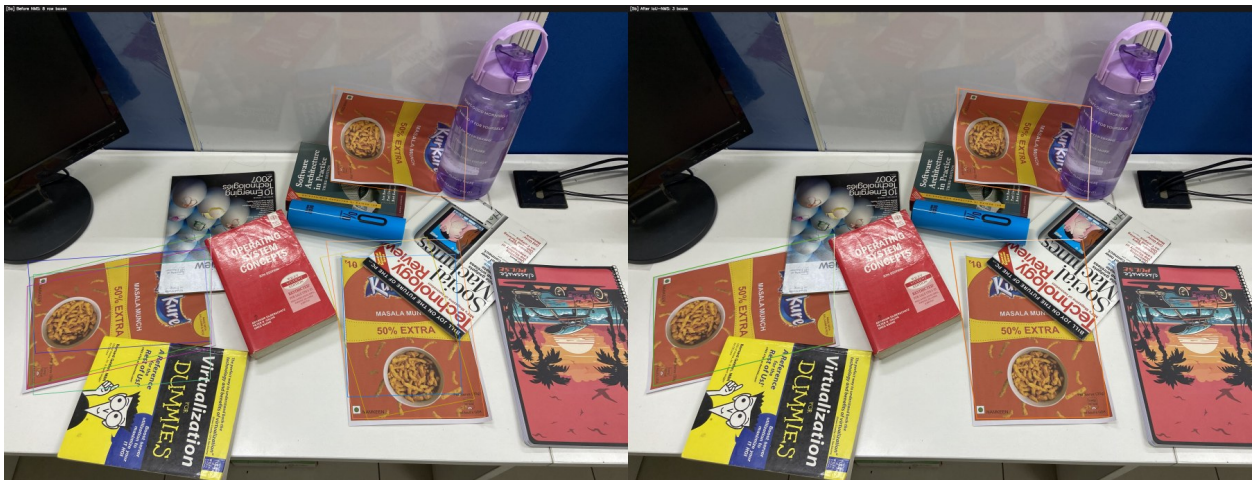


Figure 6. Duplicate suppression: overlapping hypotheses before and after IoU NMS.

4. Results

The pipeline localised **3** Kurkure packets on the lab desk. 8 raw hypotheses were produced before NMS; 5 were removed as duplicates; 295 bins failed verification. Total runtime was 50.0 s.

#	Scale	Rot (°)	Inliers	Ratio	Err (px)	ZNCC	σmax/σmin
1	0.234	92.2	450	0.87	6.45	0.14	1.107
2	0.293	90.2	594	0.92	4.81	0.39	1.358
3	0.321	359.1	874	0.85	5.88	0.23	1.154

Table 1. Accepted detections.

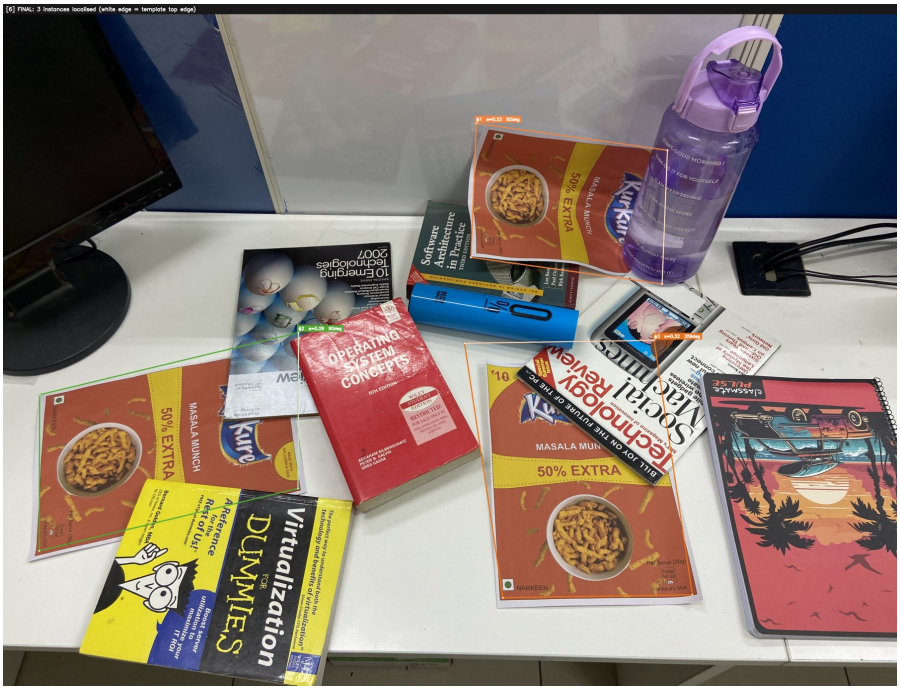


Figure 7. Final localisations. Each quadrilateral is the template rectangle projected through the estimated affine transform.



Figure 8. Comparison required by the brief: chaotic naive feature matching versus the final processed bounding-box detections.

5. Discussion

Naive matching alone is not usable in a cluttered desk scene: most correspondences attach to background texture. Lowe's ratio test removes a large fraction of ambiguous pairs, but still leaves an unordered match set. Clustering in pose space groups those matches into instance hypotheses; RANSAC and the geometric checks keep the projected boxes from shearing; verification rejects spurious peaks; and IoU NMS leaves one box per packet.

The recovered scales (about 0.23–0.32 relative to the close-up template) and rotations near 0° and 90° match the layout visible in the lab-desk photograph. Mean reprojection errors on the accepted models are a few pixels on a 4K scene.

6. How to reproduce

```
pip install -r requirements.txt
python run_pipeline.py --template data/template.png --scene data/scene.png --no-gt --out
outputs
python make_report.py
```